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Studierichting: Informatica
Oefeningenreeks 4A nr. 42.14

$$I = \int \frac{3dx}{2x^2 - 5x + 2}$$

$$D = 24 - 4 \cdot 2 \cdot 2 = 25 - 16 = 9$$

$$x_1 = \frac{1}{2}, x_2 = 2$$

$$\Rightarrow \frac{3}{(2x-1)(x-2)} = \frac{A}{2x-1} + \frac{B}{x-2}$$

$$A = \left.\frac{3}{x-2}\right|_{x=\frac{1}{2}} = -2$$

$$B = \left.\frac{3}{2x-1}\right|_{x=\frac{1}{2}} = 1$$

$$\Rightarrow I = \int \frac{-2dx}{2x-1} + \int \frac{dx}{x-2}$$

$$= -2 \cdot \frac{1}{2} \ln|2x-1| + \ln|x-2| + C = \ln|x-2| - \ln|2x-1| + C$$

References